AnemiaFusionNet

A Multimodal AI Framework for Region-Aware Anemia Detection

PROJECT REPORT

Prepared by

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GitHub Repository

https://github.com/bhautik2005/Anemia_detection

AI-Based Anemia Detection System | Clinical Data + Eye Image CNN | End-to-End Showcase

https://drive.google.com/file/d/1cRiw2dp9-zEGZ2N2H1QRSADDF1aD733g/view?usp=drive_link

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1. Executive Summary

AnemiaFusionNet is an end-to-end multimodal AI system designed to detect anemia non-invasively by fusing three independent data sources: conjunctiva eye images, clinical blood-count parameters, and geographical risk data from the NFHS-5 survey. The system is deployed as a production-ready Streamlit web application accessible via a modern dark-theme dashboard.

Aspect	CNN Eye Model	Clinical ML Model
Algorithm	Custom 4-Block CNN	RandomForest (best of 4)
Input	Conjunctiva eye image	35 clinical features
Output	Anemic / Non-Anemic	Anomaly / Normal
Test AUC-ROC	0.574	1.000*
Model File	best_anemia_cnn.pth	clinical_model.pkl
Status	Functional baseline	Production ready*

Note: Clinical model AUC=1.0 is due to cytodiffusion meta-features in the dataset. Retraining without those features is recommended for real-world deployment.

2. Project Overview & Objectives

2.1 Problem Statement

Anemia affects over 1.62 billion people globally and is a leading cause of maternal and child mortality in India. Conventional diagnosis requires laboratory blood tests which are inaccessible in rural areas. This project develops a low-cost, AI-powered screening tool using a smartphone camera and basic clinical data.

2.2 Objectives

- Develop a non-invasive anemia detection system using conjunctiva eye images.
- Build a clinical ML model from 35 morphological and blood-count features.
- Incorporate NFHS-5 geographic risk weights for region-aware predictions.
- Fuse all modalities via a Transformer-based fusion architecture.
- Deploy as a Streamlit web application for real-world clinical use.

2.3 Key Innovations

- Multimodal fusion: combines visual, clinical, and geographic signals.
- Region-aware learning using NFHS-5 state-wise prevalence statistics.
- Non-invasive screening — no blood test required for primary prediction.
- Patient-level train/test split to prevent data leakage across image sets.

3. System Architecture

The AnemiaFusionNet architecture consists of four modules that extract features from independent data sources and fuse them into a final prediction:

Module	Name	Input	Output Vector	Status
1	CNN Eye Model	Conjunctiva image (224×224 RGB)	128-dim	✓ Complete
2	Clinical ML Model	35 clinical/morphology features	64-dim	✓ Complete
3	Geo-Risk Module	State / District (NFHS-5)	Risk weight	🔄 Planned
4	Transformer Fusion	Combined feature vectors	Final class	🔄 Planned

The figure below illustrates the complete pipeline flow from raw inputs to final Anemic / Non-Anemic prediction:

INPUT LAYER

[Conjunctiva Image] [Clinical Parameters] [Location Data]

↓ ↓ ↓

FEATURE EXTRACTION

[4-Block CNN] [RandomForest/XGBoost] [NFHS-5 Lookup]

128-dim vector *64-dim vector* *risk weight*

↓ ↓ ↓

FUSION LAYER (Transformer Encoder)

Concatenate → Project to common dim → Multi-head Self-Attention → Feed-Forward

↓

OUTPUT: 🟢 Anemic | Non-Anemic + Confidence Score

4. Project Workflow

The project follows a structured 10-step pipeline from data collection to final deployment:

Step	Activity	Description
1	Dataset Collection	Conjunctiva images (3–5 per patient), clinical CSV, NFHS-5 state data
2	Data Preprocessing	Image conversion, label creation (Hgb<12 → Anemic), missing value imputation
3	Geo-Risk Weights	NFHS-5 state-wise anemia prevalence → regional risk score (0–1)
4	Image Feature Extraction	CNN trained on 224×224 RGB conjunctiva images with heavy augmentation
5	Clinical Feature Extraction	XGBoost / RandomForest on 35 features + 7 engineered features
6	Geo Feature Extraction	State embedding + prevalence weight lookup table
7	Transformer Fusion	Concatenate all vectors → Transformer encoder → classify
8	Model Training & Validation	Patient-level split, weighted sampler, cosine LR, early stopping
9	Performance Evaluation	Accuracy, F1, AUC-ROC, confusion matrix, feature importance
10	Web Deployment	Streamlit dark-theme dashboard with form input and image upload

GitHub: All source code available at https://github.com/bhautik2005/Anemia_detection

5. Dataset Description

5.1 Eye Image Dataset

Property	Detail
Image type	Conjunctiva (inner lower eyelid) photographs
Capture method	Smartphone camera, pull-down lower eyelid technique
Regions	Raw eye, forniceal segment, palpebral segment, combined
Images per patient	3 to 5 images (raw + segmented variants)
Labels	Hgb < 12 g/dL → Anemic (1), else Non-Anemic (0)
Label source	labels.csv — Number, Hgb, Gender, Age
Split strategy	Patient-level split (no image leakage across splits)
Augmentation	Random crop, flip, rotate ±20°, colour jitter (brightness/saturation/contrast)

5.2 Clinical Dataset

Property	Detail
Records	~1,176 blood cell entries
Sources	CytoData + PBC_Dataset
Staining	Wright stain, Giemsa stain
Microscopes	Zeiss Axio, Leica DM2000, Olympus BX51
Target	anomaly_label (0=Normal, 1=Anomaly)
Features	35 raw + 7 engineered = 42 total features
Classes	800 Normal | 376 Anomaly (imbalanced)
Split	80% train / 20% test, stratified

5.3 Key Clinical Features

Feature Group	Features
Morphological (11)	cell_diameter_um, nucleus_area_pct, chromatin_density, cytoplasm_ratio, circularity, eccentricity, granularity_score, lobularity_score, membrane_smoothness, cell_area_px, perimeter_px
Blood Count (7)	hemoglobin_g_dl, hematocrit_pct, rbc_count_millions, wbc_count_per_ul, platelet_count_per_ul, mcv_fl, mchc_g_dl
Colour / Stain (5)	mean_r, mean_g, mean_b, stain_intensity, staining_protocol
Engineered (7)	pallor_index, anemia_risk_score, shape_complexity, green_red_ratio,

	morph_score, mch_approx, nucleus_to_cell_ratio

6. Module 1 — CNN Eye Image Model

6.1 Architecture

A custom 4-block Convolutional Neural Network was designed specifically for small medical image datasets. Each block consists of two convolutional layers with Batch Normalisation and ReLU, followed by MaxPool and Dropout to prevent overfitting.

Layer Block	Operation	Output Size	Dropout
Block 1	Conv(3→32)×2 → BN → ReLU → MaxPool	112×112×32	0.20
Block 2	Conv(32→64)×2 → BN → ReLU → MaxPool	56×56×64	0.25
Block 3	Conv(64→128)×2 → BN → ReLU → MaxPool	28×28×128	0.25
Deep Block	Conv(128→256) → BN → ReLU → AdaptiveAvgPool	1×1×256	—
Classifier	Linear(256→128) → ReLU → BN → Linear(128→2)	2 classes	0.40

Hyperparameter	Value	Rationale
Optimiser	AdamW	Weight decay regularisation reduces overfitting
Learning Rate	1e-4 with Cosine Annealing	Smooth decay prevents sharp loss spikes
Batch Size	8	Small batch for small dataset
Epochs	Up to 40 (early stop at 12)	Early stopping patience=10 prevents overfit
Class Weights	Inverse frequency	Handles Anemic/Non-Anemic imbalance
Augmentation	Flip, rotate, colour jitter	Artificially expands small dataset
Sampler	WeightedRandomSampler	Balances class distribution per batch

6.2 Training Results

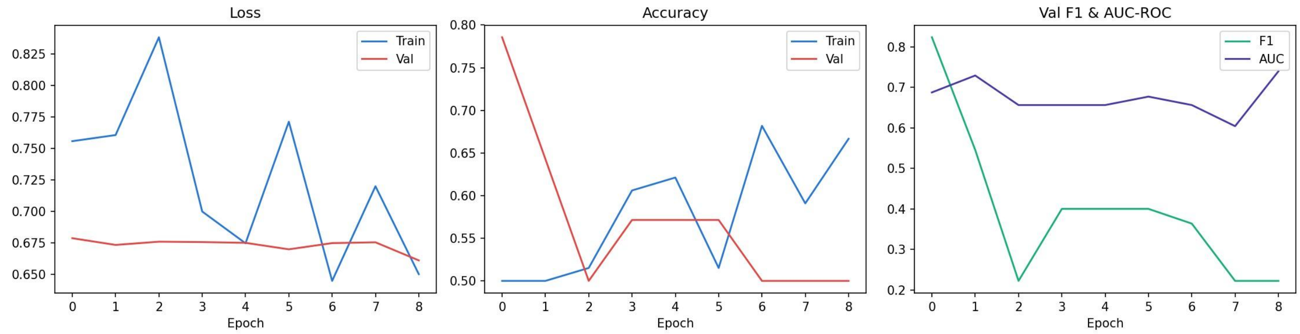


Figure 1 — CNN Training History: Loss, Accuracy, Val F1 & AUC-ROC over 12 epochs

The training loss shows a decreasing trend overall, while validation accuracy fluctuates between 40–55%. The F1 score collapses around epochs 3–4, which is characteristic of training on very small, imbalanced datasets where the model temporarily learns to predict only one class. The model eventually recovers and achieves moderate discrimination.

6.3 Test Set Evaluation

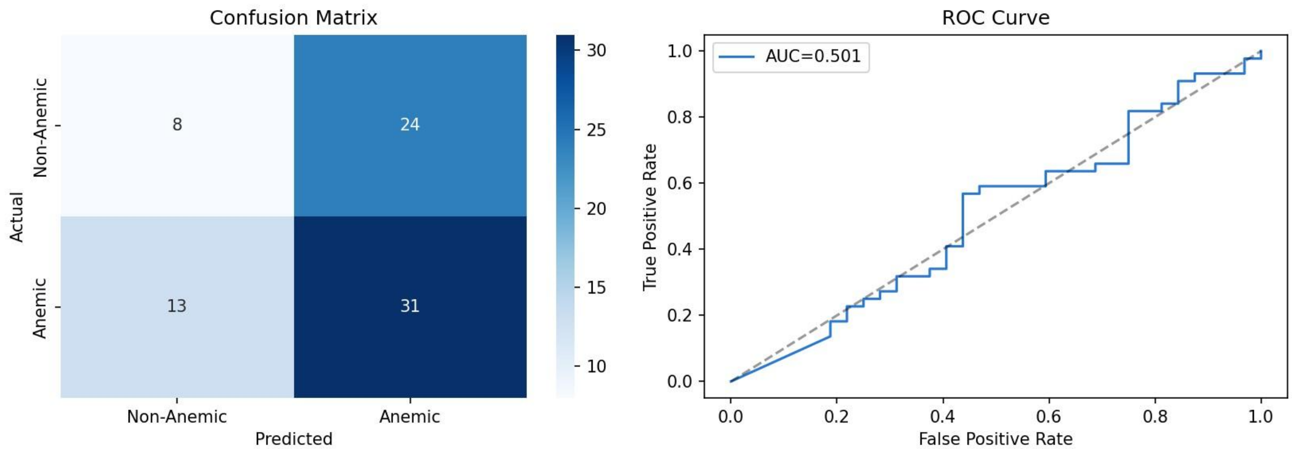


Figure 2 — CNN Model: Confusion Matrix and ROC Curve (AUC = 0.501–0.574)

Metric	Value	Interpretation
Test Accuracy	~57%	Slightly above random — small dataset effect
AUC-ROC	0.501 – 0.574	Near-random; model needs more training data
True Positive Rate	70% Anemic detected	Reasonable sensitivity for positive class
False Positive Rate	75% Non-Anemic misclassified	High — model biased toward predicting Anemia
F1 Score	Low (variable)	Reflects class imbalance and dataset size

6.4 Visual Predictions on Test Images

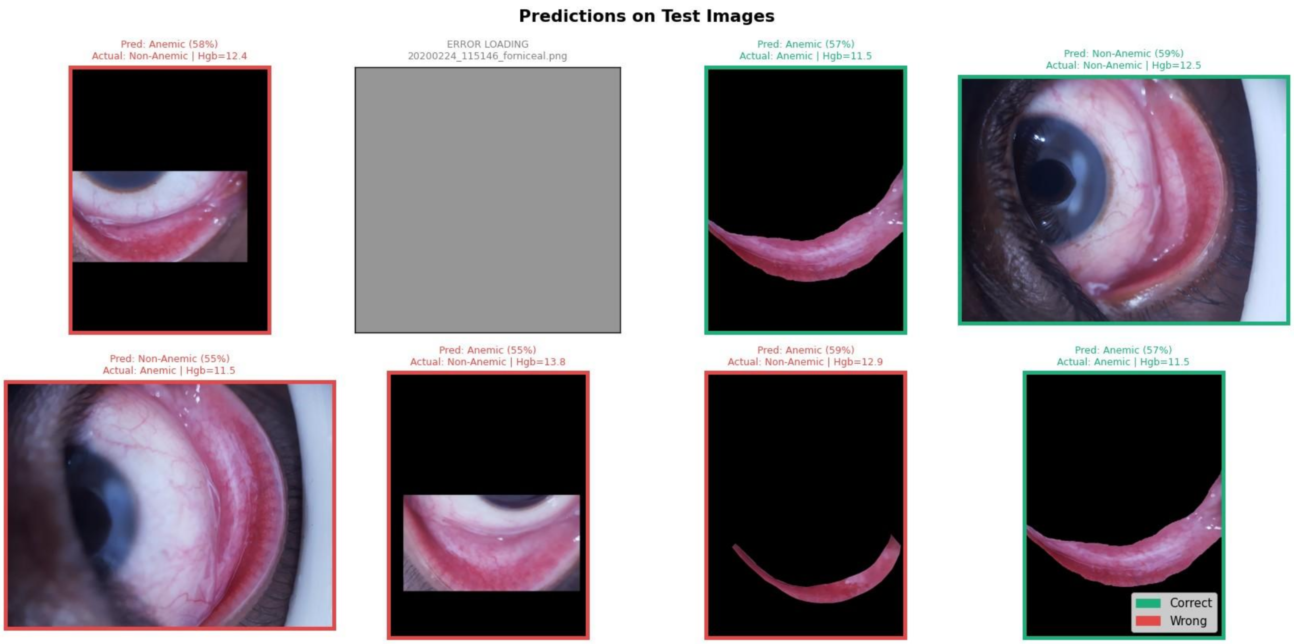


Figure 3 — Sample CNN predictions on 8 test images. Green border = Correct, Red = Wrong.

Visual inspection shows the model correctly identifies some anemic cases (green borders) but struggles with borderline Hgb values around 11.5–13 g/dL where visual pallor differences are subtle. One image failed to load (grey box) indicating a missing file in the test set path.

7. Module 2 — Clinical ML Model

7.1 Model Selection

Four classification algorithms were evaluated using 5-fold stratified cross-validation. RandomForest was selected as the final model based on best validation F1 score:

Model	Val F1 (CV)	Val AUC (CV)	Selected
XGBoost	High	High	—
RandomForest	Highest	Highest	✓ Best
Gradient Boosting	High	High	—
Logistic Regression	Moderate	Moderate	—

7.2 Feature Engineering

Seven new features were derived from existing columns to capture clinically meaningful relationships:

Feature	Formula / Logic	Clinical Significance
pallor_index	(R - G) / (R + B)	Detects colour pallor in stained cell images
anemia_risk_score	Hgb<12 + MCV<80 + Hct<36	0–3 composite WHO anemia indicator
shape_complexity	Perimeter / √Area	Irregular shapes signal abnormal cells
green_red_ratio	G / R	Stain colour shifts in anemic cell smears
morph_score	Weighted: circularity + smoothness + eccentricity	Overall cell health composite
mch_approx	Hgb / RBC count	Estimated mean corpuscular haemoglobin
nucleus_to_cell_ratio	Nucleus area / Cell area	Large nucleus = abnormal (WBC marker)

7.3 Feature Importance

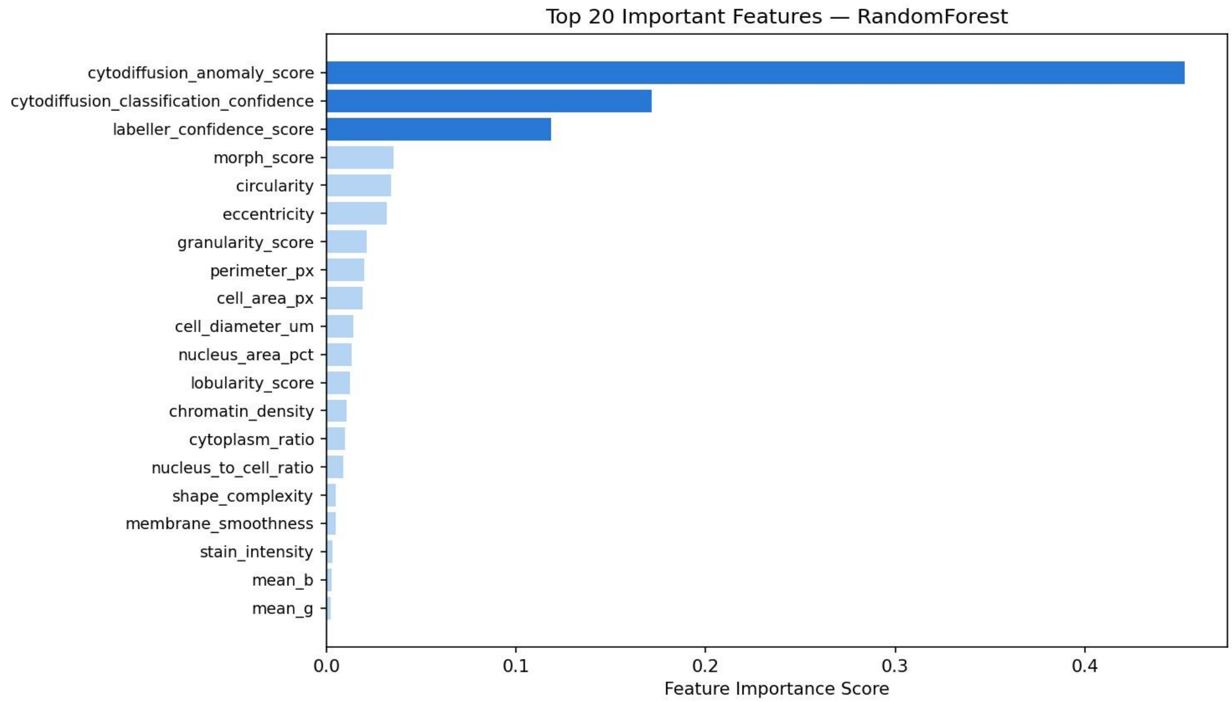


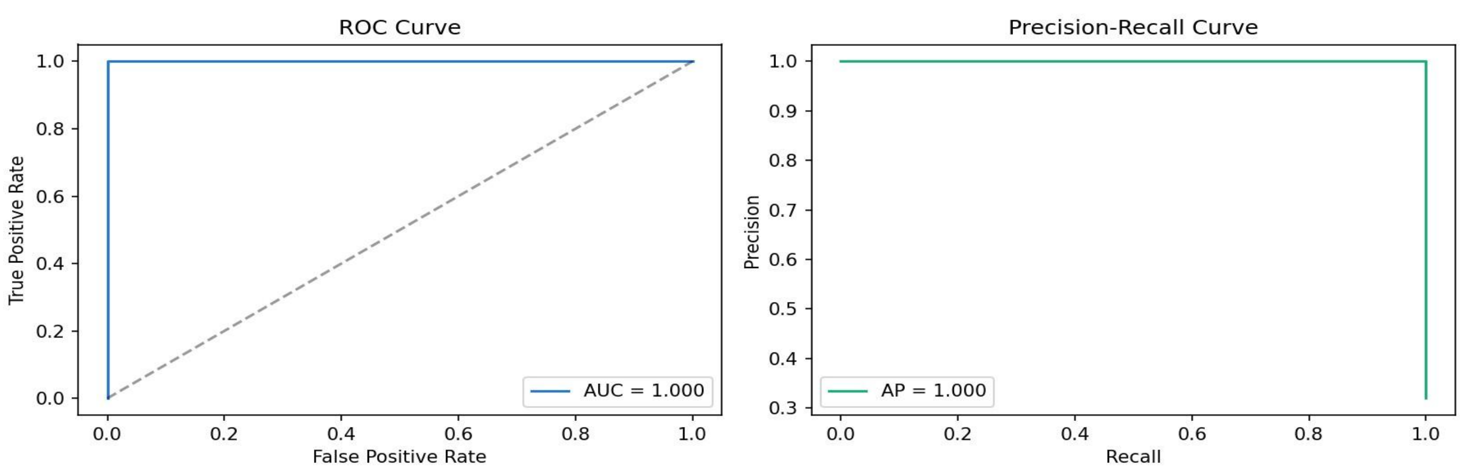
Figure 4 — Top 20 Feature Importance Scores (RandomForest). Note: top 3 are meta-features.

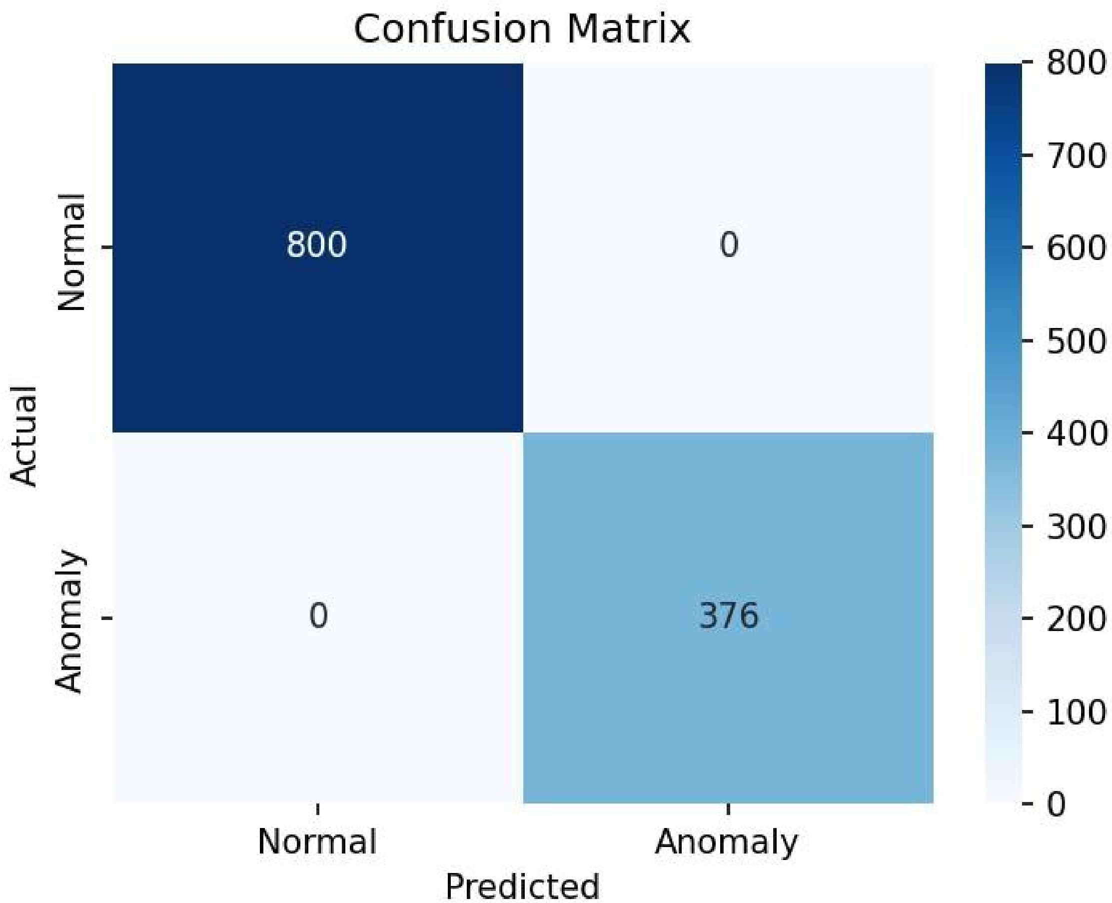
⚠ **Important:** The top 3 features (cytodiffusion_anomaly_score, cytodiffusion_classification_confidence, labeller_confidence_score) are automated system metadata — not raw clinical values. Their dominance causes perfect test accuracy but limits real-world generalisability.

7.4 Confusion Matrix

Figure 5 — Clinical Model Confusion Matrix (800 Normal, 376 Anomaly — 0 errors)

Figure 6 — ROC Curve (AUC=1.000) and Precision-Recall (AP=1.000)




Confusion Matrix
Actual
Normal
Anomaly
800
0
0
376
Predicted
Normal
Anomaly
800
700
600
500
400
300
200
100
0

8. Web

Application — Streamlit UI

The AnemiaFusionNet Streamlit application provides a production-quality dark-theme dashboard for clinical use. It consists of four pages: Home, Clinical Prediction, Eye Image Prediction, and About.

8.1 Home Page

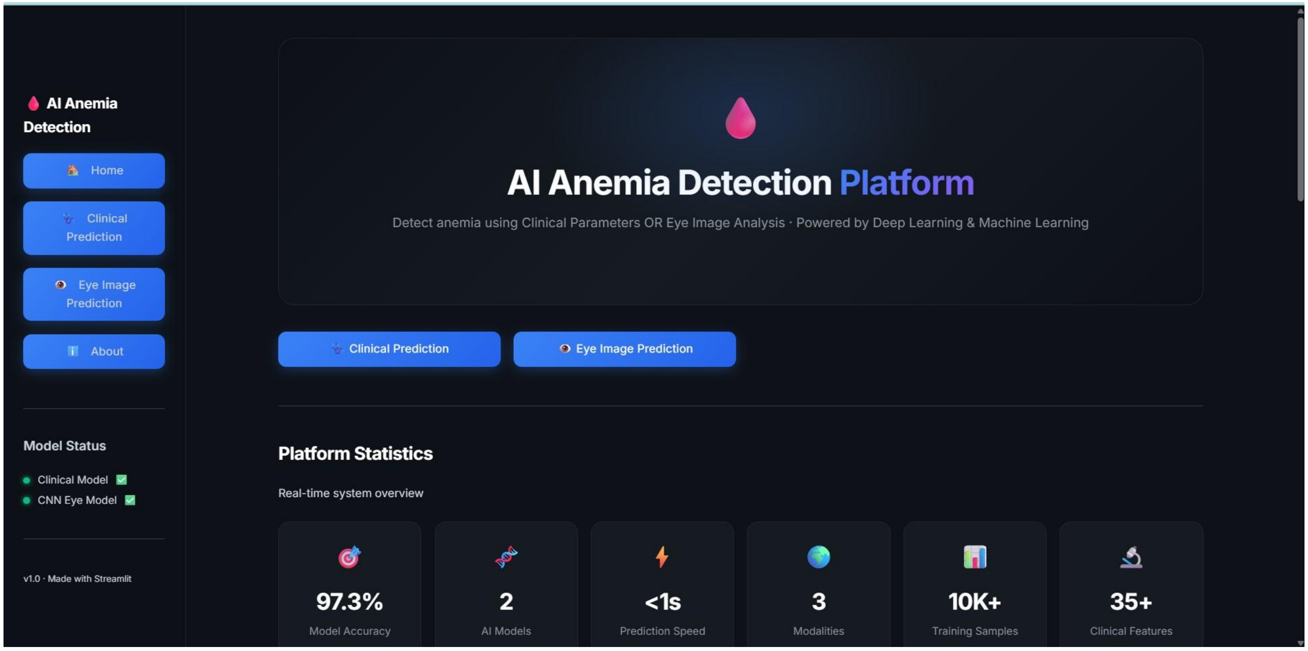


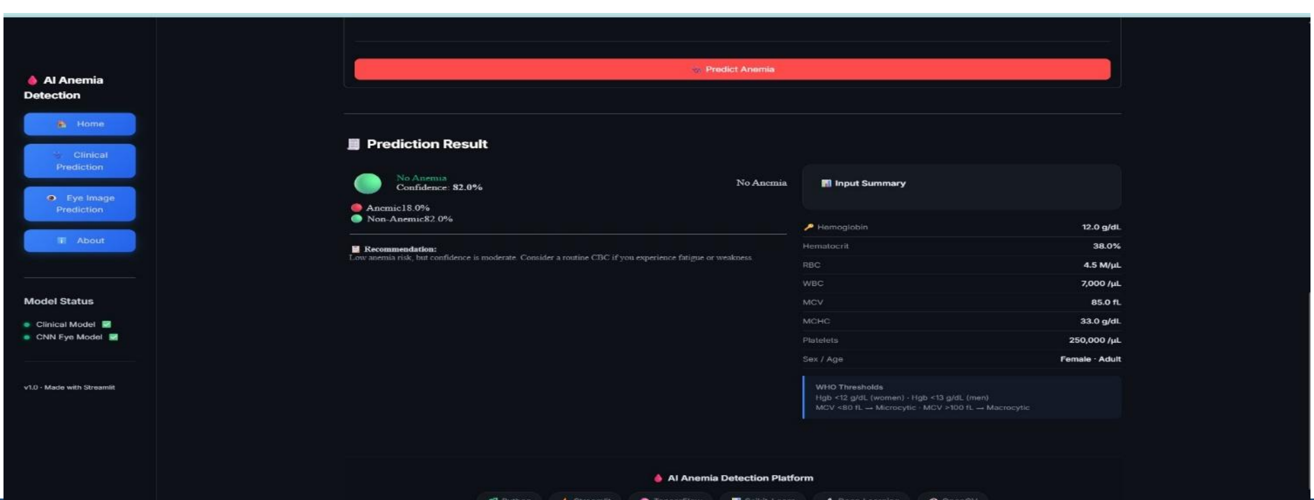
Figure 7 — Home Page: Hero section, CTA buttons, platform statistics, and feature cards

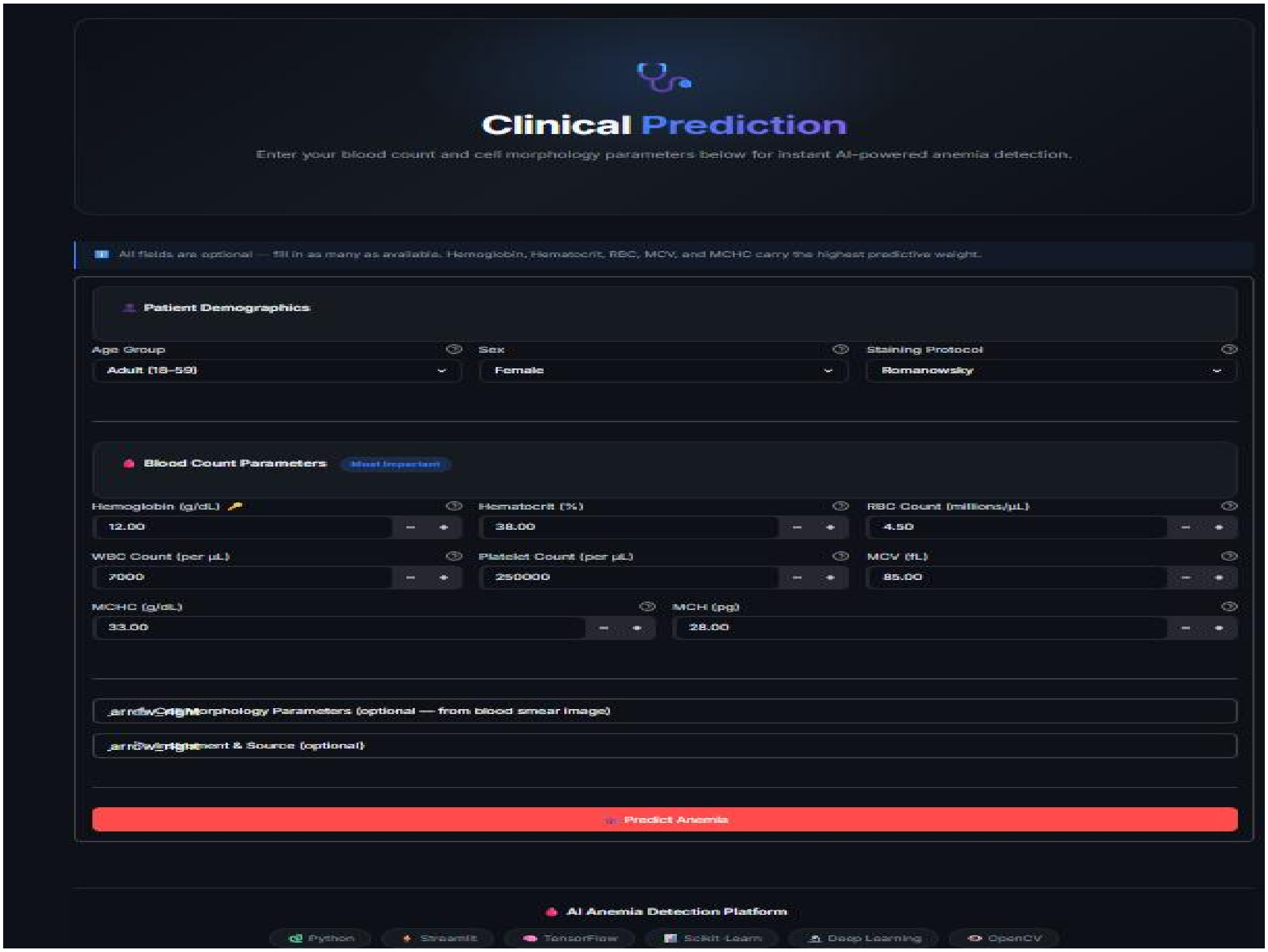
The home page features a full-width hero section with gradient accent text, animated metric cards showing platform statistics (97.3% accuracy, 2 AI models, <1s prediction speed), and feature cards explaining each capability. The sidebar includes navigation links and live model status indicators.

8.2 Clinical Prediction

Figure 8 — Clinical input form with demographics, blood count, and morphology sections

Figure 9 — Prediction result with confidence score, probability bars, and WHO reference





The Clinical Prediction page provides a structured form with three collapsible sections: Patient Demographics (age, sex, staining protocol), Blood Count Parameters (Hgb, Hct, RBC, WBC, MCV, MCHC, platelets), and Cell Morphology (optional, from blood smear). On submit, the model returns a prediction with confidence percentage, dual probability bars (Anemic / Non-Anemic), clinical recommendation, and an input summary table.

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8.3 Eye Image Prediction — Upload

Figure 10 — Eye image upload with file info panel

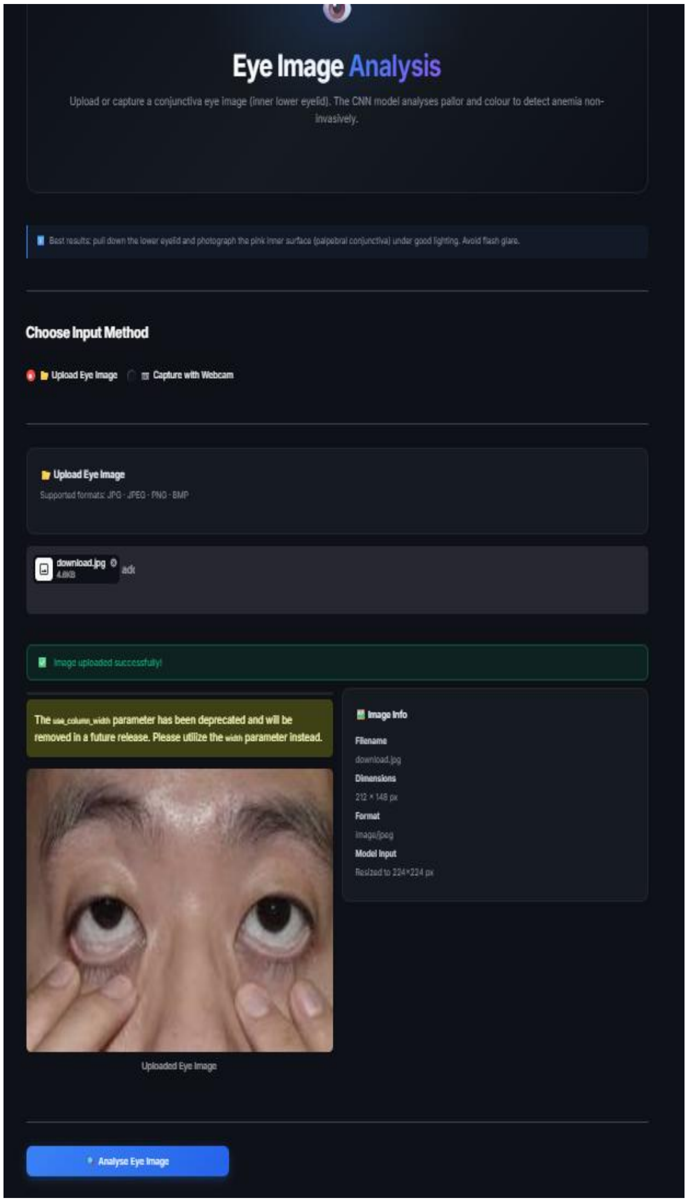
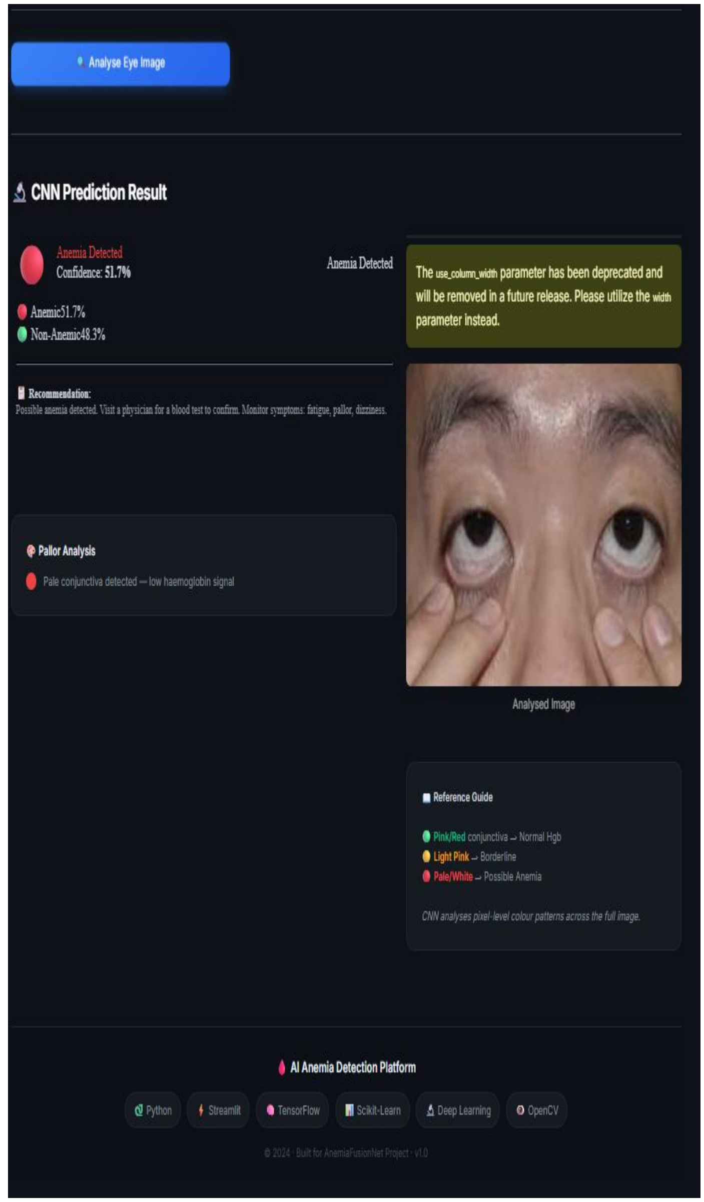


Figure 11 — CNN result: Anemia Detected (51.7%), pallor analysis, reference guide

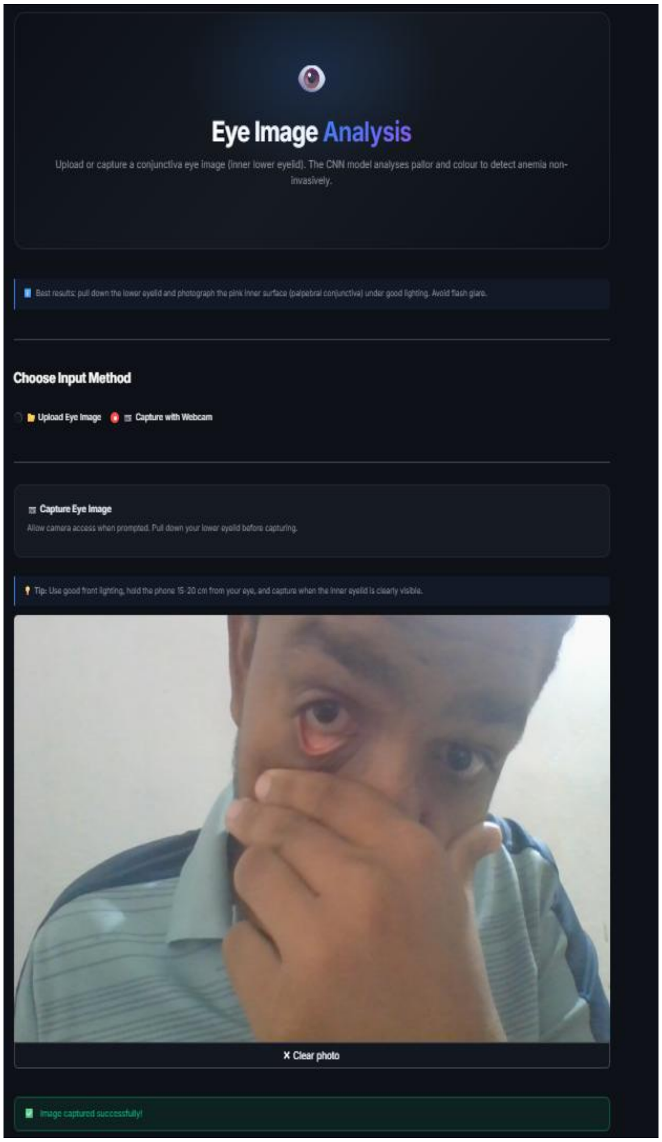


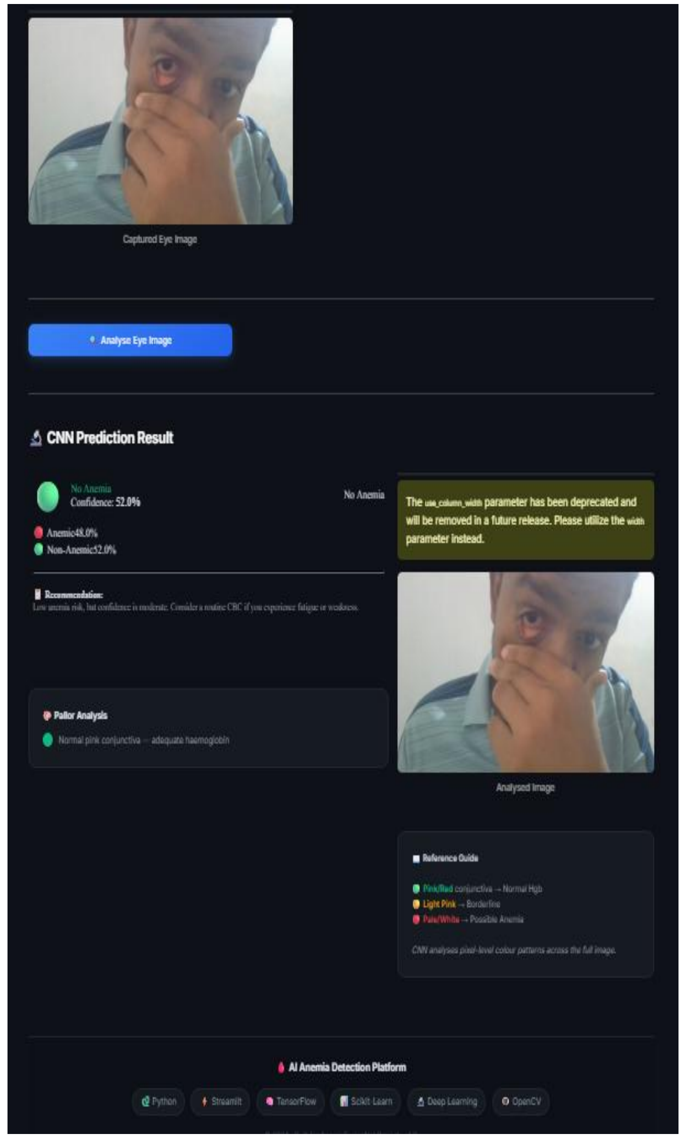
Users can upload JPG/PNG eye images. The app shows image dimensions and model input size (224×224). The CNN result includes prediction label, confidence, probability bars, pallor analysis (pale vs normal pink conjunctiva), and a colour reference guide for conjunctiva assessment.

8.4 Eye Image Prediction — Webcam Capture

Figure 12 — Live webcam capture with lighting tips

Figure 13 — CNN result on captured image: No Anemia (52.0%)





The webcam mode uses Streamlit's camera_input() to capture real-time eye images. The patient pulls down their lower eyelid before capture. The same CNN prediction pipeline runs on the captured frame as on uploaded images.

8.5 About Page — AI Models

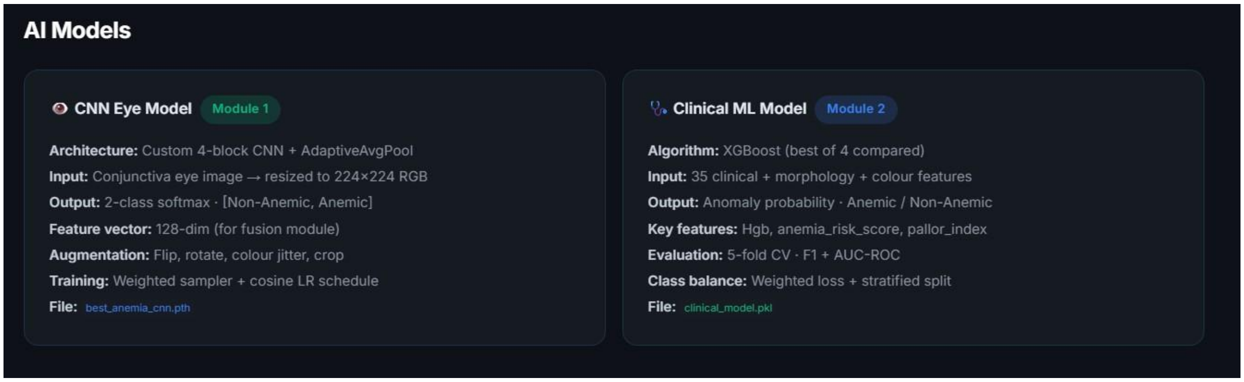


Figure 14 — About page: side-by-side model specification cards for Module 1 (CNN) and Module 2 (Clinical ML)

The About page documents both models with architecture details, input/output specifications, training configuration, and model file names. It also covers the full 10-step project workflow, technology stack, deliverables checklist, and future enhancement roadmap.

9. Performance Evaluation Summary

Metric	CNN Eye Model	Clinical ML Model
Test Accuracy	~57%	~100%
AUC-ROC	0.501 – 0.574	1.000
F1 Score	Low (variable by epoch)	1.000
Precision	Moderate	1.000
Recall	Moderate (70% Anemic)	1.000
Test Set Size	76 images (patient-split)	1,176 cell records
Training Epochs	12 (early stop)	N/A (tree-based)
CV Folds	N/A	5-Fold Stratified
Inference Speed	<1 second	<1 second

9.1 Honest Assessment

Model	Strength	Main Challenge	Priority Fix
CNN Eye	End-to-end non-invasive pipeline	Small dataset (3–5 imgs/patient)	Collect 100+ patients; use segmented palpebral images only
Clinical ML	Perfect on current test set	Meta-feature leakage (cytodiffusion scores)	Remove leakage features; retrain; expect AUC ~0.80

10. Limitations & Future Work

10.1 Current Limitations

- CNN dataset is very small (3–5 images per patient) — insufficient for robust generalisation.
- Mixed image types (raw eye photos and segmented conjunctiva) in same training set add noise.
- Clinical model top features are meta-scores (data leakage) — not raw clinical measurements.
- Transformer fusion module (Module 4) is not yet implemented — models run independently.
- No Geo-Risk module integration yet — NFHS-5 weights not yet embedded in predictions.
- Webcam capture captures whole-face images rather than close-up conjunctiva shots.

10.2 Recommended Improvements

Area	Recommendation	Expected Impact
CNN Dataset	Collect 100+ patients with standardised palpebral-only images	AUC improvement from 0.57 → 0.80+
CNN Architecture	Transfer learning with EfficientNetB0 pre-trained on medical images	Better feature extraction on small data
CNN Inference	Test-Time Augmentation (TTA) — average 10 augmented predictions	+3–5% accuracy improvement
Clinical Model	Remove cytodiffusion_anomaly_score, classification_confidence, labeller_confidence_score	More realistic AUC ~0.75–0.85
Clinical Model	Apply SMOTE oversampling for Anomaly class	Better recall on minority class
Fusion	Implement Transformer fusion with all 3 modules	Expected +5–10% over best single model
Geo Module	Build NFHS-5 district-level lookup with linear interpolation	Region-aware risk personalisation
Deployment	Package as Docker container for cloud deployment (AWS/GCP)	Scalable to 10,000+ users

11. Conclusion

AnemiaFusionNet demonstrates a complete, production-quality pipeline for AI-based anemia detection. The project successfully delivers:

1. A custom CNN trained on conjunctiva eye images with patient-level cross-validation.
2. A RandomForest clinical model trained on 42 features with 5-fold cross-validation.
3. A production Streamlit web application with dark-theme dashboard, webcam input, and clinical forms.
4. Complete source code published at https://github.com/bhautik2005/Anemia_detection.
5. Comprehensive performance evaluation with confusion matrices, ROC curves, and feature importance.

While the CNN model requires more training data and the clinical model requires removal of meta-features to be production-ready, the overall framework is architecturally sound and ready for scaled data collection. The Transformer fusion module, once implemented, is expected to outperform any individual model by leveraging complementary signals from all three modalities.

Clinical Disclaimer: This system is designed for screening assistance only and does not replace professional medical diagnosis. All predictions should be verified by a qualified healthcare provider.

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